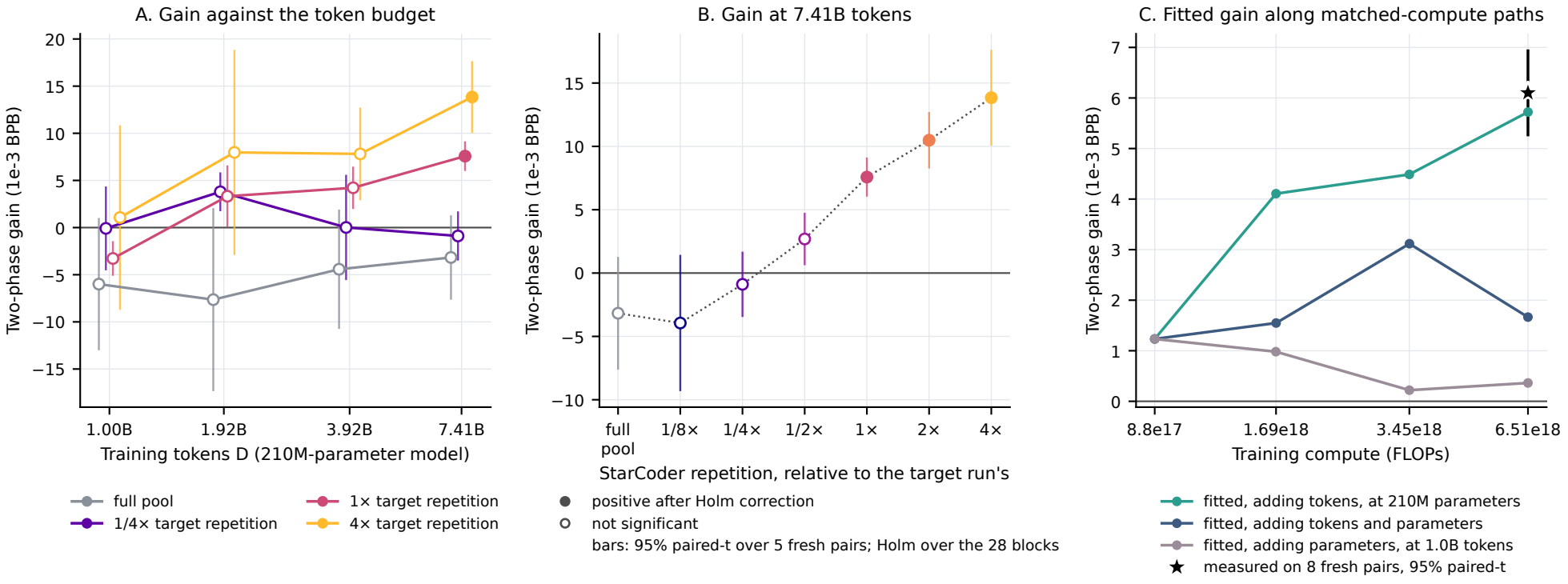
A. Gain against the token budget
Two-phase gain (1e-3 BPB)
20
15
10
5
0
−5
−10
−15
1.00B
1.92B
3.92B
7.41B
Training tokens D (210M-parameter model)
B. Gain at 7.41B tokens
Two-phase gain (1e-3 BPB)
15
10
5
0
−5
−10
full pool
1/8×
1/4×
1/2×
1×
2×
4×
StarCoder repetition, relative to the target run's
C. Fitted gain along matched-compute paths
Two-phase gain (1e-3 BPB)
7
6
5
4
3
2
1
0
8.8e17
1.69e18
3.45e18
6.51e18
Training compute (FLOPs)
full pool
1/4× target repetition
1× target repetition
4× target repetition
positive after Holm correction
not significant
bars: 95% paired-t over 5 fresh pairs; Holm over the 28 blocks
fitted, adding tokens, at 210M parameters
fitted, adding tokens and parameters
fitted, adding parameters, at 1.0B tokens
measured on 8 fresh pairs, 95% paired-t